DEEPA LOKESHA

Embedded Systems / Hardware Engineer | Embedded C/C++ • PCB Design (Altium) • Hardware Bring-Up & Debug

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EDUCATION

University of Pennsylvania — Philadelphia, PA

Master of Science in Engineering (MSE), Electrical Engineering | Aug 2025 – May 2027 | GPA: 3.30/4.00

REVA University — Bangalore, India

Bachelor of Technology, Electronics and Communication Engineering | Dec 2020 – Jun 2024 | GPA: 8.77/10.00

TECHNICAL SKILLS

Embedded & Firmware: Embedded C, C/C++, FreeRTOS, Zephyr / nRF Connect SDK, Real-Time Systems, Register-Level Firmware, Secure FOTA (MCUboot/OTA)

PCB & Hardware Design: Altium Designer, Schematic Capture, 4-Layer PCB Design, Power Management (LiPo/Buck-Boost), Hardware Bring-Up & Debugging, LabVIEW, MPLAB

Protocols & MCUs: SPI, I2C, UART, MQTT, CoAP/DTLS, BLE, Wi-Fi, ADC/DAC, GPIO · ARM Cortex-M, SiWG917, ESP32, nRF7002/5340, ATmega328

Cloud, Tools & AI: nRF Cloud, Memfault, Node-RED, Azure, Git, VS Code, Python, MATLAB, PID Control, Sensor Fusion, ML Fundamentals, Oscilloscope & Thermal-Camera Diagnostics

EXPERIENCE

Accenture Solutions Pvt Ltd — Jul 2024 – Jun 2025

Packaged App Development Associate — *Bangalore, India*

- Completed 340+ hours of backend engineering training (Java, Spring Boot, RESTful APIs, Agile/DevOps); applied Git, SDLC practices, and systematic debugging across cross-functional delivery teams.

Vikram Sarabhai Space Centre (ISRO) — Aug 2023 – Sep 2023

Project Intern — *Thiruvananthapuram, India*

- Architected an FPGA-based FIR filter (Verilog/VHDL) to eliminate 1–2 kHz beat-frequency interference in capacitance-to-voltage propellant-level sensors on semi-cryogenic launch vehicles; validated in MATLAB and Libero SoC with a 4-member team.

Bharat Electronics Limited (BEL) — Jul – Aug 2023

Intern — Electronic Warfare & Avionics — *Bangalore, India*

- Observed end-to-end PCB fabrication (chemical etching, drilling, lamination, surface finishing) on high-precision industrial equipment, with exposure to radar, avionics, and signal-processing systems.

PROJECTS

RunFast — Custom-PCB IoT Gait-Analysis Wearable — Jan – May 2026

- Designed the complete 4-layer PCB in Altium (schematic capture, component placement, routing) around a Silicon Labs SiWG917 (Cortex-M4 + Wi-Fi), an LSM6DSVETR IMU (I2C), dual FSR sensors via an AD8606 front end, and BQ24075 + TPS62082 power.
- Assembled and soldered the full hardware stack, validated IMU and dual-FSR sensor performance through bench testing, and built the Node-RED/Azure MQTT dashboard for live stride-metric visualization and session logging.
- Diagnosed and recovered a hardware bring-up failure with zero board respins — isolated an excess current draw to a swapped op-amp VCC/GND footprint using an oscilloscope and thermal camera, then fixed it with trace cuts and jumper wires.

DrySpool — Connected Filament-Storage System — Sep 2026 – Present

- Building a Nordic-based connected filament-storage system (nRF7002 DK, Zephyr / nRF Connect SDK) in a 3-person team — environmental sensing with dehumidification, NFC spool ID, and a shared makerspace dashboard.
- Team implementation on the nRF5340 (TF-M secure / non-secure split) adds Wi-Fi telemetry to nRF Cloud (CoAP/DTLS), Memfault diagnostics, and a secure signed OTA pipeline (MCUboot, ECDSA-P256) with a verified two-version update and rollback.

Smart Chair — Posture & Occupancy Monitoring System — Nov – Dec 2025

- Built a Bluetooth-connected posture and occupancy monitor from thin-film pressure, PIR, and ToF sensors — hand-soldered the full sensor array and implemented sensor-fusion logic to improve occupancy-detection accuracy.

Additional Projects

- **Autonomous Differential-Drive Robot (Penn) —** Owned hardware integration on a dual-ESP32 robot in a 4-person team: hand-crimped Molex harnesses, soldered GPIO for Vive, IMU, and motor drivers, and tuned PID wall-following.
- **Streamline Your Drive (REVA) —** Developed a smart-parking and EV/CNG-station monitoring system using IR, ultrasonic, RFID, and GSM/GPS sensors on Arduino/ESP32.
- **Autonomous Robotic Vehicle (REVA) —** Built an ATmega328-based obstacle-avoidance robot in embedded C for real-time autonomous navigation across a 2–400 cm sensing range.

LEADERSHIP, CERTIFICATIONS & HONORS

- **Co-President, ESE Graduate Association (ESEGA) —** University of Pennsylvania (2026–Present).
- Altium Designer Essentials — Altium Education (Jul 2026) · Tech Expressway Program — Accenture (Nov 2024).
- Recognized by Altium Education for outstanding project work — University of Pennsylvania Demo Day 2026.